DILTIAZEM

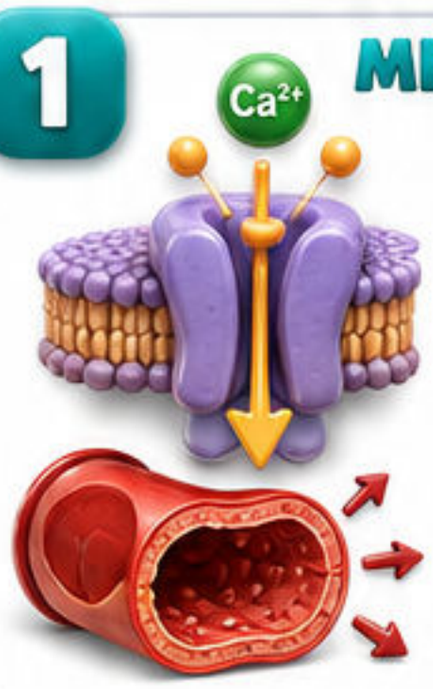
[Cardizem, Tiazac]

Class: Antianginal | Non-Dihydropyridine Calcium Channel Blocker



Pharma Talk

1 MECHANISM OF ACTION (MOA) & TARGET



- **TARGET:** L-type calcium channels
- **MOA:** Inhibits calcium influx during depolarization ➡ ⬇ vascular smooth muscle tone ➡ systemic vasodilation.
  ⬇ sinoatrial and atrioventricular node conduction velocity ➡ ⬇ heart rate and myocardial oxygen demand.
- **Analogy:** Electrical brake and vascular relaxer.

2 FORMS & STRENGTHS



- **Tablet:** 30, 60, 90, 120 mg
- **Extended Release:** 120–420 mg
- **Injection:** 5 mg/mL

3 INDICATIONS (FDA-APPROVED & MAJOR OFF-LABEL)



- **FDA-APPROVED:**
  - ✔ Hypertension
  - ✔ Chronic stable angina
  - ✔ Vasospastic angina
  - ✔ Atrial fibrillation ventricular rate control
- **MAJOR OFF-LABEL:**
  - ✔ Atrial ectopic tachycardia
  - ✔ Pulmonary arterial hypertension
  - ✔ Esophageal spasm

4 DOSAGE & ADMINISTRATION

A ADULT DOSE

- **Standard:** 120–360 mg PO daily
- **Route:** Oral / IV
- **Frequency:** Once daily (ER) or divided (IR)
- **Max:** 540 mg daily
- **Loading/Maintenance:** N/A

B) PEDIATRIC DOSE

PEDIATRIC USE — Safety/efficacy not established.

C) ADMINISTRATION



- IV Bolus: 0.25 mg/kg over 2 minutes
- May repeat: 0.35 mg/kg after 15 minutes
- Continuous Infusion: 5–15 mg/hour
- **Route:** Oral / IV

D) ADJUSTMENT



- **Renal:** Use caution
- **Hepatic:** ⬇ dose required (Child Pugh B/C)

5 PHARMACOKINETICS & DYNAMICS (PK/PD)



- **Onset (IV):** 2–5 minutes
- **Peak Effect (IV):** 5–15 minutes
- **$t_{1/2}$:** 3–4.5 hours
- **Metabolism:** Hepatic CYP450
- **Excretion:** Feces (60%), Urine (~35%)

6 ADVERSE EFFECTS & TOXICITY



- **Common:** Peripheral edema, headache, dizziness, bradycardia, constipation, flushing, fatigue.
- **Severe:** Heart block, cardiogenic shock, hepatotoxicity, severe hypotension, worsening heart failure.



7 CONTRAINDICATIONS & INTERACTIONS



- **Absolute CI:** Heart failure with reduced ejection fraction, sick sinus syndrome, 2nd/3rd-degree AV block, severe hypotension
- **Boxed Warning:** None
- **Major Interaction:** Simvastatin ➡ CYP450 inhibition ➡ ⬆ myopathy risk.



8 SPECIAL POPULATIONS (PREGNANCY & LACTATION)



- **Pregnancy:** Category C
- **Lactation:** Excreted; avoid
- **Geriatric:** Start low; titrate slowly and monitor
- **Pediatrics:** Safety/efficacy not established



9 CLINICAL PEARLS



- ✔ Never combine with IV beta-blockers ➡ risk of profound bradycardia and heart block.
- ✔ Max simvastatin dose is 10 mg daily.
- ✔ Avoid in HFrEF; safe in HFpEF.
- ✔ Monitor ECG and blood pressure continuously during IV infusion for AF.
- ✔ Always check baseline liver function tests.

10 GLOSSARY OF ABBREVIATIONS



- **CYP450** = Cytochrome P450
- **ECG** = Electrocardiogram
- **FDA** = Food and Drug Administration